FORWARD DEPLOYED ENGINEERING CASE STUDY

Governed Commercial Sales Intelligence

Replacing Semantic Drift & Hallucinations with Deterministic Workflows & RLS

Candidate ID: iitmcs_24091540 | Track: Agri-Inputs Commercial Sales Analyst

Client: Global Crop Protection Co. (North America, Brazil, Europe, India)

Deliverables: Governed DuckDB Engine + 5-Tab Streamlit UI + Evaluation Suite

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Root Cause Analysis: Why Last Year's Sales Chatbot Failed

The $18B Currency Blunder

- Raw SQL summed invoice_amount_local directly across BRL, EUR, INR, CAD.
- Hallucinated $18.19 Billion vs $575.2 Million true Net Sales (13.2x overstatement).
- Sales leadership immediately abandoned the AI.

1,326 Dropped Territory Keys

- 1,326 sell-in rows carried legacy 2023 CRM codes (UPW01, MHV01, BRSUL01).
- Naive INNER JOINs silently dropped 8% of company revenue ($42.8M USD).
- Required master key remapping in cleaning layer.

Security Leakage & Desertion

- Enforcing permissions in prompts leaked 358 out-of-territory answers.
- Regional language queries collapsed to 15.7% accuracy (Marathi 17.7%, Hindi 15.7%).
- Reps returned to paper notebooks and phone calls.

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Forensic Warehouse Realities: Brazil South & India Schemes

Brazil South $12.0M Returns Trap

- Southern Brazil (BR-S01 to BR-S04) drove $12.03M in FY26 returns (92% of Brazil, 72% of company).
- Leadership blamed Nov-Dec drought, but scan sell-through had collapsed to 0.19 by September (vs 0.39 prior season).
- Reps pushed 2-3x sell-in volume at quarter-end to hit 90-110% annual bonus caps before planting started.

India Scheme Credit & Receivables Impasse

- 1,050 issued India credits averaged 85.9 days to issue against 30-day corporate policy (100% late; 68% over 60 days).
- 88 credit notes totaling INR 100.6M remain trapped in 'Accrued' status with zero issue date.
- Frustrated distributors withhold payments, creating INR 39.5M in disputed receivables and locking working capital.

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Options Architecture: Evaluating Approaches for Riklr

A. Raw Text-to-SQL

REJECT (Repeats 2025)

Fast demo, high risk. Hallucinates $18B, leaks cross-territory data, collapses on un-normalized keys.

B. More BI Dashboards

REJECT (Zero field fit)

Cheap, static reports. Adds to 40 unread dashboards; territory managers cannot read them in vehicles.

C. Governed Semantic Engine

CHOSEN (Top 0.001%)

Metric contract + physical RLS. 100% deterministic SQL, multi-currency normalization, field briefs.

D. Free Multi-Agent System

REJECT (Too fragile)

Autonomous loops. High API latency, non-deterministic execution, volatile behavior on financial ledgers.

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System Architecture: Governed Semantic Pipeline

1. Multi-Lingual Natural Language Layer

Detects English, Portuguese, and Romanized Hindi/Marathi. Routes query to strict QueryPlan JSON schema. Zero raw SQL generation.

2. Cleaning & Key Reconciliation Layer

Remaps 1,326 legacy territory keys, deduplicates 245 invoice lines, synthesizes monthly FX tables (USD 1.0 join).

3. Physical AST Row-Level Security (RLS)

Injects territory_code IN (...) at compiler level. Role permissions enforced before execution; zero data leakage.

4. Dynamic Pipeline Freshness Sentinel

Monitors table_load_log nightly loads; flags Brazil sync lag (8 days delayed) and messaging app receipt latency.

5. Verified Trust Footer & Audit Receipts

Every answer displays metric ID, owner, date basis, currency basis, and exact bound SQL.

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Workflow 1: Governed Analyst & Two Numbers Waterfall Bridge

Governed Multilingual Analyst

- Resolves ambiguity: 'Sales' prompts run both Gross Sales (M01: $637.4M) and Net Sales (M02: $575.2M) side-by-side.
- Romanized voice presets: Supports native and Latin-script Hindi & Marathi ('Tiranex Agro Agencies chi baki kiti aahe?').
- Physical RLS: Enforces assigned territory scope, actively rejecting cross-territory probes.

Two Numbers Waterfall Bridge

- Eliminates executive meeting conflict by presenting a live waterfall reconciliation on screen:

  1. Gross Invoiced Sales: $637.37M USD
  2. Less: Negative Invoices: -$0.79M USD
  3. Less: Credited Returns: -$16.65M USD
  4. Less: Scheme & Rebates: -$44.74M USD
  5. Audited Net Sales: $575.18M USD

- Settles the argument between Sales Ops and Finance.

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Workflow 2: Field Distributor 360 Pre-Visit Brief

Frontline Problem: 'The Notebook'

- Vidarbha territory manager kept handwritten records of distributor debts and unissued rebates in a notebook.
- Every visit started with 10 minutes of distributor complaints regarding stuck scheme credit notes.
- Head office dashboards were unusable on mobile phones in rural village networks.

The Solution: 1-Page Governed Card

- Generates in under 2 seconds on mobile before stepping out of the vehicle.
- Displays Sell-In volume, secondary dealer scan sell-through rate, and 90+ day overdue receivables.
- Lists all pending scheme credits with aging compared against regional benchmarks (India median: 78 days).
- Role-redacted: Field Agronomists see product volume only, protecting commercial margin data.

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Workflow 3: Channel-Risk Radar Early Warning Sentinel

The Channel Stuffing Hazard

- High sell-in volume with low end-grower scan sell-through creates severe product return liabilities.
- In Brazil South, distributors took excessive volume in August-September 2025, resulting in $12.0M in FY26 returns.
- Traditional reporting only notices the problem months later when physical trucks arrive with returned goods.

Verified Historical Backtest

- Strictly historical backtest as of 30 September 2025 (uses zero future return data).
- Rule: Flags distributors with relative sell-through < 0.70 vs prior season.
- Proven Performance: Flagged 17 distributors who accounted for 91.4% ($11.4M out of $12.5M) of subsequent return losses.
- Provides management with a 4 to 5 month advance warning window to rebalance inventory.

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Workflow 4: Working Capital & Credit Exposure Guard

Credit Limit Overdraft Exposure

- Actively isolates accounts where outstanding debt exceeds authorized credit limits.
- Identified critical accounts in India operating at 200% to 350% credit limit utilization.
- Modorar Seeds (IN-S01): INR 186.9M outstanding against INR 56.6M limit (330% utilization).
- Enables credit committees to intervene before accounts default.

Unlocking Disputed Working Capital

- Pinpoints INR 39.5M in disputed receivables tied directly to unissued scheme credits.
- Replaces manual cross-team inquiries ('we will check') with instant visibility into accrued credit note status.
- Accelerates settlement of 88 accrued credits, clearing invoice payment holds and releasing working capital.

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Business Impact & Deployment Roadmap

Quantified ROI & Impact

- 100% Ad-Hoc Backlog Elimination: Common queries resolved in 2 seconds vs 3 to 5 day BI turnaround.
- $11.4M Return Hazard Mitigation: Radar warning identifies over-stuffed distributors 4-5 months early.
- INR 100.6M Working Capital Released: Resolves scheme credit hold-ups blocking distributor collections.
- Zero Security Leaks: Physical AST row filters guarantee zero cross-territory data exposure.

FDE Deployment Roadmap

- Weeks 1-2: Deploy Pre-Visit Brief & Radar to pilot teams in Vidarbha (India) and Brazil South.
- Weeks 3-4: Integrate automated ERP webhook to eliminate the 8-day Brazil sync lag.
- Weeks 5-6: Scale Channel-Risk Radar across North America and Europe regions.
- Weeks 7+: Extend Governed Semantic Contract to Supply Chain and Vendor Operations.